<u>Assignment Sheet – II</u>

Course: BCA	**Year/Semester:** 2^{nd} / 4^{th}	**Session:** 2024-25
Subject Name & Code: BCAC 0023: PROGRAMMING IN JAVA		

1. What is **Inheritance**? Explain all types of inheritance in Java with examples of each.
2. What is the purpose of the **super** keyword in Java? How is the super() constructor used in a child class? Why must it be the first statement in the constructor? Can super and this be used together in the same constructor? Explain with an example.
3. What is the difference between a **static** variable and a final variable in Java? How does marking a method as final affect method overriding? What happens if a final method is inherited?
4. What is the use of **this** keyword in java. Explain with example.
5. What is the use of **static** keyword in java. Explain with example.
6. Explain the concept of method **overloading** in Java. Explain with example.
7. Explain the concept of method **overriding** in Java. Explain with example.
8. What are the different **access** specifiers in Java, and what is their scope? Explain the difference between protected and default access specifiers with examples. Can a private method of a parent class be overridden in a subclass? Why or why not?
9. What is an **abstract** class, and how is it different from a concrete class? Can an abstract class have a constructor? If yes, provide an example. Explain why we cannot create an object of an abstract class, but we can have its reference.
10. What is an **interface**? How is an interface different from an abstract class in Java? How does Java 8 enhance interfaces with default and static methods? Provide examples. Can an interface extend another interface? Can a class implement multiple interfaces? Explain with examples.
11. What is a **package** in Java, and why is it used? How do you create and access classes in custom packages? How can you prevent package members from being accessed by classes outside the package?
12. What is the difference between checked and unchecked **exceptions** in Java? How does the finally block differ from the catch block in exception handling? Explain how custom exceptions are created and used in Java with an example.
13. What is **multithreading**? How is it achieved in Java (1) By extending thread (2) By implementing Runnable interface. Also explain thread priorities in Java.

Programming Problems:

14. Create a base class Employee with properties like name and salary. Derive a class Manager that adds properties like teamSize and a method to calculate bonus based on team size. Create another subclass Intern with additional properties like collegeName and a method to calculate stipend. Demonstrate the functionality by creating objects of each class.

15. Create a class Account with an overloaded deposit() method to handle deposits for cash, cheque, and online transactions. Override the withdraw() method in a subclass SavingsAccount to add specific rules like minimum balance.
16. Create an abstract class Shape with an abstract method calculateArea(). Extend it with Circle, Rectangle, and Triangle classes to implement the method.
17. Create an interface RemoteControl with methods like turnOn(), turnOff(), and changeChannel(). Implement this interface in TV and AirConditioner classes.
18. Write a program to take two numbers as input and divide them, handling ArithmeticException for division by zero.
19. Create a custom exception InvalidAgeException and throw it when the user inputs an age below 18.
20. Create a program with two threads: one to print even numbers and the other to print odd numbers up to a given limit.

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